SC7 Bayes Methods MT26

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Lecture 4: Rewards, decisions and utility

Probability as preference

Student: *What is subjective probability?*

Teacher: It expresses the strengths of preferences you hold for different sets of outcomes.

Student: *What is preference?*

Teacher: Your order $A \succ B$ express the belief that $\theta \in A$ is more likely to hold than $\theta \in B$.

Student: *What do you mean by "more likely to hold"?*

Teacher: It's subjectively more probable.

Student: *Aren't we trying to define subjective probability?*

Resolution: preferences $A \succ B$ satisfying SA can be represented as probabilities and "more likely to hold" means $\pi(A) > \pi(B)$.

The Axioms of Probability just define algebraic properties of a probability space.

The Frequentist definition of probability (proportion of successes in an infinite sequence of trials) defines the real-world data distribution $p^*(\cdot)$ for our problem, if it exists.

It is our job to make an observation model $p(\cdot|\theta)$ that satisfies the AoP and approximates p^*.

The Savage Axioms just define the algebraic properties of preference relations. If preference relations satisfy SA then a unique prior expressing the preferences and satisfying the AoP exists.

de Finetti's Theorem shows that if $Y_1, Y_2, \ldots$ is an IES there is a real generative model with $Q \sim F$ and $Y_1, Y_2, \ldots | Q \sim Q$ iid.

It is our job to write down a set of preference relations satisfying the SA and giving $\pi(\cdot)$ approximating F.

Decision Theory in the context of parameter estimation

SA ensure the whole framework of decision theory is available.

For an action $a \in \mathcal{D}$ and truth $\theta \in \Omega$ our loss function is $L(\theta, a)$. We seek a decision rule δ which determines the action we take.

The loss is elicited, like the prior. We ask the scientist for their objective. Suppose it is parameter estimation.

For data $y \sim p(y|\theta),\ y \in \mathcal{Y}$ the action is an estimate $\hat{\theta} = \delta(y)$ and the rule is an estimator $\delta : \mathcal{Y} \to \mathcal{D}$ with risk

$$\mathcal{R}(\theta, \delta) = \int_{\mathcal{Y}} L(\theta, \delta(y)) p(y|\theta) dy.$$

If we have a prior $\pi(\theta)$, posterior $\pi(\theta|y)$ and marginal likelihood $p(y)$ the expected posterior loss (EPL) is

$$\rho(\pi, a|y) = \int_{\Omega} L(\theta, a) \pi(\theta|y) d\theta.$$

The Bayes risk $\rho(\pi, \delta) = E_{\theta\sim\pi}(\mathcal{R}(\theta, \delta))$ is closely related,

$$\rho(\pi, \delta) = \int_\Omega \int_\mathcal{Y} L(\theta, \delta(y))p(y|\theta)\pi(\theta)dyd\theta,$$

since if $p(y)$ is the marginal likelihood, then $\rho(\pi, \delta)$ is

$$\rho(\pi, \delta) = \int_\mathcal{Y} \rho(\pi, \delta(y)|y)p(y)dy.$$

The Bayes estimator δ^π for θ,

$$\delta^\pi = \arg\min_\delta \rho(\pi, \delta)$$

is given by minimising the EPL at each $y \in \mathcal{Y}$,

$$\delta^\pi(y) = \arg\min_{a\in\mathcal{D}} \rho(\pi, a|y).$$

Example: if $\theta \in \Omega$ is discrete and $L(\theta, a) = \mathbb{I}_{\theta\neq a}$ so we seek the true θ-value, and all else is equally bad, $\rho(\pi, \theta|y) = 1 - \pi(\theta|y)$ is the EPL and our estimator $\delta(y) = \arg\max_{\theta\in\Omega} \pi(\theta|y)$ is the mode.

Actions, rewards, utility and loss:

Let $r(\theta, a)$ be the reward for action a and truth θ.

Let $U(r),\ r_{\min} \leq r \leq r_{\max}$ denote the utility of reward r.

Utility is the opposite of loss. In terms of our previous notation,

$$L(\theta, a) = \text{const} - U(r(\theta, a)),$$

replacing one function L by two, U and r, which must be elicited.

At fixed $y \in \mathcal{Y}$ and $\theta \sim \pi(\cdot|y)$ the random reward $R = r(\theta, \delta(y))$ for the action $a = \delta(y)$ has distribution $R \sim P_{\delta,y}(\cdot)$.

Changing the rule δ changes $P_{\delta,y}$. Choose δ to maximise $E(U(R))$:

$$U(R) = U(r(\theta, \delta(y))), \quad \theta \sim \pi(\cdot|y)$$

$$E_{\theta|y}(U(r(\theta, \delta(y)))) = \text{const} - E_{\theta|y}(L(\theta, \delta(y))).$$

At fixed y, maximise posterior expected utility $E(U(R))$ to minimise EPL. A rule maximising at each y minimises Bayes risk.

Coherent prior belief - the objective is to order sets

Consider two sets, $A, B \subseteq \Omega$, action space $a \in \{A, B\}$ reward $r(\theta, a) = \mathbb{I}_{\theta \in a}$ and utility $U(0) = 0,\ U(1) = u$ for $u > 0$.

There is no data here so the rule is a fixed action and $a = \delta$. The random reward $R \sim P_\delta$ is defined by $R = r(\theta, \delta),\ \theta \sim \pi(\cdot)$.

The reward distribution, $P_\delta = (P_\delta(0), P_\delta(1))$, for $\delta = B$ is

$$P_\delta = (1 - \pi(B), \pi(B))$$

The expected utility $E_\theta(U(r(\theta, \delta)))$ of the rule $\delta = B$ is

$$P_\delta(0)U(0) + P_\delta(1)U(1) = u\pi(B)$$

so we maximise $E(U)$ by choosing B if $\pi(B) > \pi(A)$ etc.

For this to work we need the prior π and utility U to exist.

Let $\pi(A) > \pi(B) \Rightarrow A \succ B$ define a preference order $(\mathcal{B}_\Omega, \succeq)$.*

The idea here is that we assume a true generative process

$$\psi \sim F,\ y \sim p^*(\cdot|\psi)$$

exists (for example the data are an IES).

Our preferences $(\mathcal{B}_\Omega, \succeq)$ *are our prior*.

In our model for the true generative process $\psi \in \Omega$ and if $A \succ B$ then we think $F(A) > F(B)$ so we want $\pi(A) > \pi(B)$.

If we start with $(\mathcal{B}_\Omega, \succeq)$ then we seek a prior expressing it,

$$\pi(A) > \pi(B) \quad \Leftarrow \quad A \succ B.$$

If this prior exists and is unique we have coherent belief.

*$(\mathcal{B}_\Omega, \succeq)$ is a binary relation over $A, B \in \mathcal{B}_\Omega$ with $A \sim A$ and if $A \succeq B$ then $A \succ B$ or $A \sim B$. Also, $A \succ B \Leftrightarrow B \prec A$.

Coherent posterior belief

We ultimately need a posterior $\pi(\cdot|y)$ to exist for each $y \in \mathcal{Y}$.

We can then go forward with the reward distribution $R \sim P_{\delta,y}$ defined by $R = r(\theta, \delta(y)),\ \theta \sim \pi(\cdot|y)$.

However, we simply assume the observation model $p(\cdot|\theta)$ exists. If the prior exists and $0 < p(y) < \infty$ then the posterior

$$\pi(\theta|y) = \frac{p(y|\theta)\pi(\theta)}{p(y)}$$

is proper and clearly exists.

This time we don't start with a set of preferences, we get them from the posterior

$$\pi(A|y) > \pi(B|y) \quad \Rightarrow \quad A \succ B.$$

So we focus on existence of the prior.

The Ellsberg paradox This paradox shows that peoples' preferences are in some cases inconsistent with any prior.

Two urns, (A) has 50 red and 50 black balls, (B) has 100, numbers unknown. Bet $£1000$ on the color of draw. For each bet pick an option.

	Option 1 $\text{color}_{\text{urn}}$	Option 2 $\text{color}_{\text{urn}}$
Bet 1	r_A	b_A
Bet 2	r_B	b_B
Bet 3	r_A	b_B
Bet 4	r_B	b_A

We are indifferent on Bet 1/2 if "red" and "black" are exchangeable.

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We are indifferent on Bet 1/2 if "red" and "black" are exchangeable. However, many people prefer fixed odds (urn A) to subjective uncertainty (urn B), taking r_A/b_A in Bet 3/4 respectively.

Our preferences for each bet tell us about our unstated prior $\pi(\phi)$ for the probability $\phi = \Pr(b_B)$ to draw black from urn B.

Suppose rewards have utilities $U(-1000) = 0$ and $U(1000) = u$, with $u > 0$. Indifference on Bet 1 is reasonable as

$$E(U|\text{choose } r_A) = E(U|\text{choose } b_A) = u/2.$$

On Bet 2, $E(U|\text{choose } r_B) = u(1 - E_\pi(\phi))$ and

$$E(U|\text{choose } b_B) = uE_\pi(\phi),$$

so indifference here implies $E_\pi(\phi) = 1/2$ in our prior.

On Bet 3, choosing r_A over b_B gives $E_\pi(\phi) < 1/2$. Choosing b_A over r_B in Bet 4 gives $E_\pi(\phi) > 1/2$, a contradiction and so a prior does not exist.

Expected utility hypothesis

Assume coherent prior belief, so $r(\theta, \delta) \sim P_\delta$ exists for $\delta \in \mathcal{D}$.

Consider the order $\succeq_U$ on P_δ and $P_{\delta'}$ given by

$$P_\delta \succeq_U P_{\delta'} \quad \Leftarrow \quad E_{P_\delta}(U(R)) \geq E_{P_{\delta'}}(U(R))$$

If we *start* with a preference over reward distributions then we need a utility function $U(r)$ expressing the order

$$P_\delta \succeq_U P_{\delta'} \quad \Rightarrow \quad E_{P_\delta}(U(R)) \geq E_{P_{\delta'}}(U(R))$$

in order to have coherent inference.

The expected utility hypothesis (EUH) says there exists a prior π and a utility function U such that our order over actions corresponds to order by expected utility.

Coherent inference chooses the action to maximise the expected utility. It is possible if the EUH holds.

The Allais paradox Our preferences may be inconsistent with any utility function. In this example the reward distributions are given, so the "prior" exists.

Let $p = (p_1, p_2, p_3)$ be the probability you win respectively

$$(r_1, r_2, r_3) = (£0, £500,000, £750,000).$$

In each of two rounds you have a choice between two lotteries.

1. (A) with $p^{(A)} = (0, 1, 0)$ *OR* (B) with $p^{(B)} = (0.01, 0.89, 0.1)$
2. (C) with $p^{(C)} = (0.89, 0.11, 0)$ *OR* (D) with $p^{(D)} = (0.9, 0, 0.1)$

The Allais paradox Our preferences may be inconsistent with any utility function. In this example the reward distributions are given, so the "prior" exists.

Let $p = (p_1, p_2, p_3)$ be the probability you win respectively

$$(r_1, r_2, r_3) = (£0, £500,000, £750,000).$$

Consider two separate choice problems, with the same initial wealth. In each case you have a choice between two lotteries.

1. (A) with $p^{(A)} = (0, 1, 0)$ *OR* (B) with $p^{(B)} = (0.01, 0.89, 0.1)$
2. (C) with $p^{(C)} = (0.89, 0.11, 0)$ *OR* (D) with $p^{(D)} = (0.9, 0, 0.1)$

People comonly choose (A) and (D). Choices (B) and (D) maximise the expected return. What utility function are they using?

Set the utilities to be $U(r_1) = 0$, $U(r_2) = u$ and $U(r_3) = 1$.

The expected utilities are $E(U) = (p_1, p_2, p_3)(0, u, 1)^T$, so

$$\begin{aligned} E(U|A) &= u & \qquad E(U|B) &= 0.1 + 0.89u \\ E(U|C) &= 0.11u & \qquad E(U|D) &= 0.1 \end{aligned}$$

Preferring (A) to (B) means

$$u > 0.1 + 0.89u \quad \Rightarrow \quad u > 10/11.$$

On the other hand preferring (D) to (C) means

$$0.1 > 0.11u \quad \Rightarrow \quad u < 10/11,$$

which is a contradiction. This paradox shows that human decision making does not always maximise an expected utility. This is unsurprising. However, choosing (A),(D) does seem reasonable.